



FADO

A Reaction Controller for Fairness-Aware Online Learning Under Concept Drift

Tiago Silva

CISUC · Data-Centric AI Lab

Department of Informatics Engineering, University of Coimbra



Outline

1. Motivation & Problem
2. FADO: Base Learner
3. The Reaction Controller
4. Experimental Design
5. Results
6. Intersectional Fairness
7. Takeaways & Next Steps

1. Motivation & Problem

Concept drift degrades fairness, not only accuracy

Fairness-aware learners are trained assuming a **stationary** distribution.

Real streams drift: $P(y | x)$ and base rates move with policy, behavioural and economic change.

Drift breaks the statistical relationships fairness rests on — so **bias can grow while accuracy looks stable**.

RQ1

To what extent does drift affect group-fairness metrics online?

RQ2

Can adaptive detection & response preserve fairness while keeping performance?

Keep the in-processing fairness; add the drift machinery

In-processing fair

e.g. Aranyani, RFR

Fair during training
No drift detection
Regulariser is static

Drift-adaptive

e.g. ARF

Detect & react to shift
Accuracy-driven only
No fairness mechanism

Drift-aware fair

prior attempts

Shift as fixed perturbation
Worst-case, not streaming
No online controller



FADO = retain Aranyani's node-level fairness regulariser, then wrap it in an online detector + reaction controller.

Aranyani is inherited, not contributed — the novelty is the drift-detection-and-response layer.

2. FADO: Base Learner

Why a **soft-routed oblique** forest

Oblique nodes split on a **linear combination of features** (an angled hyperplane) — far more expressive per node than axis-aligned splits.

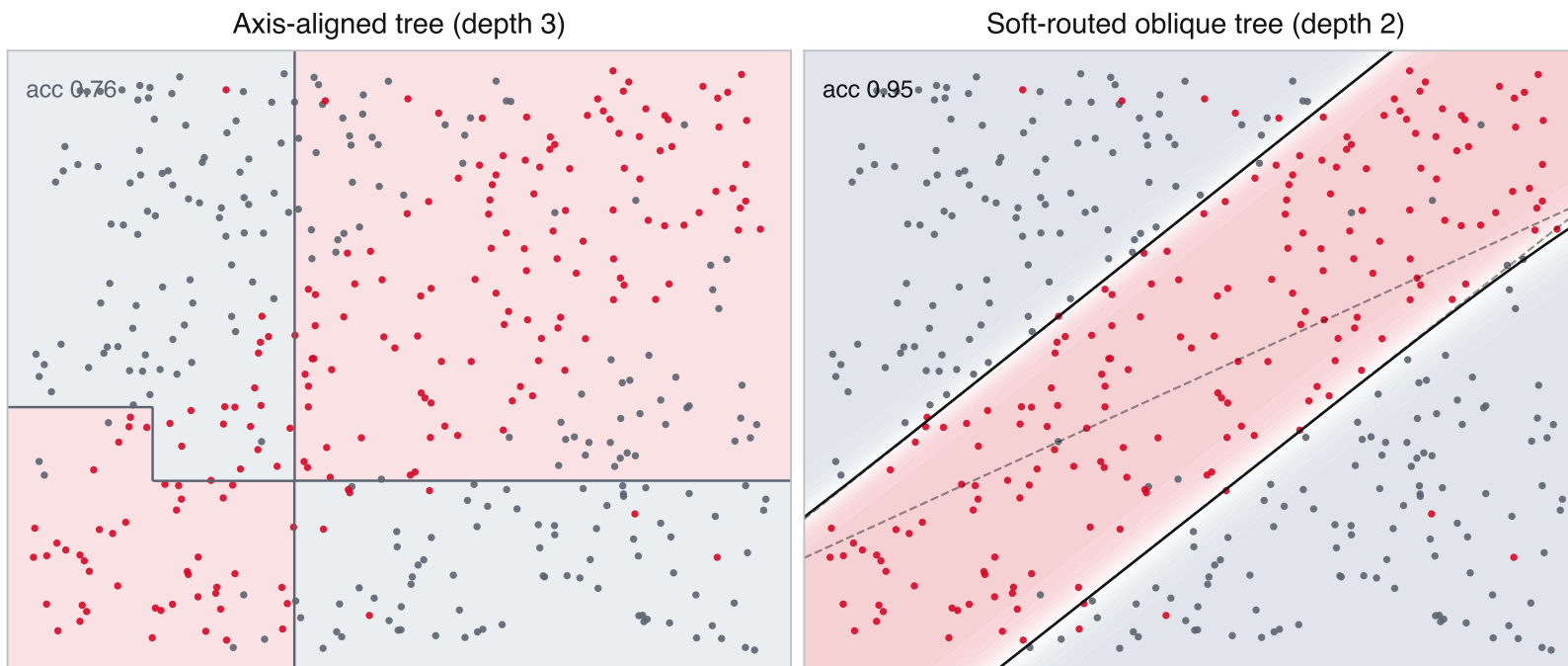
A temperature-scaled **sigmoid** makes routing soft and differentiable, so the whole forest trains by gradient descent, online — and each node acts as a per-step prediction, letting fairness be imposed at every node.

$$n_j(x) = \sigma((w_j^\top x + b_j) / \tau) \in (0, 1)$$

$$f(x) = \frac{1}{K} \sum_k \text{softmax}(f_k(x))$$

Parameter isolation $\rightarrow$ cheap per-step gradient $\cdot$ $O(W)$ streaming memory $\cdot$ **temperature τ is the controller's lever** ($\tau=1$ smooth learning; $\tau \rightarrow 0$ decisive recovery).

Oblique + soft routing, learned by **backpropagation**



Same diagonal-stripe data: axis-parallel splits approximate it with a **staircase**; two **oblique** hyperplanes capture the band, and the shading is the **soft** routing probability. (A depth-2 soft oblique tree, trained here by gradient descent.)

$$p(x) = \sum_{\ell} P_{\ell}(x) s_{\ell}, \quad P_{\ell}(x) = \prod_{j \in \text{path}(\ell)} g_j \text{ or } (1 - g_j), \quad g_j = \sigma(w_j^{\top} x + b_j)$$

Every gate is a differentiable sigmoid, so $p(x)$ is smooth in every weight — **trainable by backpropagation, one sample at a time**; the same gates g_j carry the fairness penalty and the temperature τ .

A per-node DP gap, smoothed by a **Huber surrogate**

At each routing node, compare the group-conditional mean routing to the cross-group mean — the per-node analogue of the demographic-parity gap — and sum the penalty over **every node in every tree**.

$$F_a = \bar{n} - \bar{n}_a$$

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \sum_{\text{nodes}} \frac{1}{K} \sum_a \phi_\delta(F_a)$$

$$\phi'_\delta(z) = z \text{ if } |z| < \delta, \quad \text{sign}(z) \text{ if } |z| \geq \delta$$

Quadratic near 0 (smooth gradient) · linear tail (bounded outlier influence) · **unit-slope tail** → **firmer correction past δ** . DP defined as deviation-from-mean to match the regulariser exactly.

3. The Reaction Controller

Two detectors, three phases, one **temperature**

WARN

pre-warm

Sensitive D_w trips
 $\text{LR} \times \beta_w (=5)$
arm — don't commit

CONFIRM

→ *spike + sharpen*

Strict D_c confirms
 $\text{LR} \times \beta_c (=10)$
 $\tau \rightarrow \tau_{\text{drift}}=0.1$
reset detectors

RECOVER

→ *anneal back*

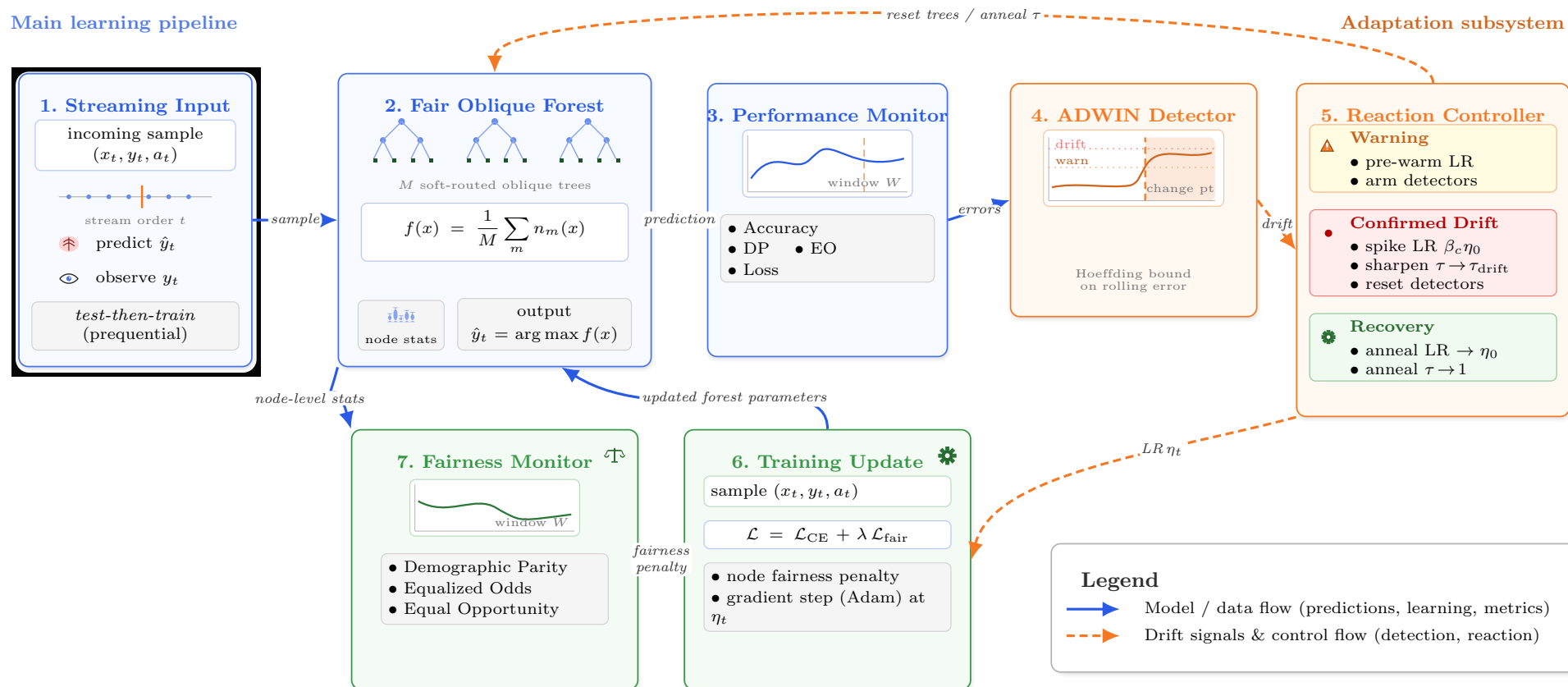
$\eta \rightarrow \eta_0$ over Δ
 $\tau \rightarrow 1$
cooldown $C=200$

$$\varepsilon_{\text{cut}} = \sqrt{\frac{1}{2m} \ln \frac{4}{\delta}}$$

δ **calibrated to stream length** ($\varepsilon \propto 1/\sqrt{m}$). Label-noise guard ($\nu=0.7$) excludes confident-but-wrong points, so noise can't fire the controller.

The prequential loop

Main learning pipeline



Blue = data flow through the fair forest. Orange = drift signals & control flow: detector $\rightarrow$ controller $\rightarrow$ retuned gradient step.

4. Experimental Design

Two complementary **drift regimes**

Folktables — natural, gradual

ACS Income (CA), ~187k rows
Real cross-year shift
2015 train → 2017–18 stream
Group = sex × race (joint)

COMPAS — synthetic, abrupt

7.2k rows, controlled testbed
Symmetric race swap +
Bernoulli(0.5) label flip
no_drift vs abrupt_race

Protocol

Prequential test-then-train
Batch 1 · 30 pinned seeds
Paired Wilcoxon + Holm
Welch t-test cross-check

Baselines (held fixed): ARF (adaptive, unfair), RFR (fair, static), and a controller-free Aranyani ablation sharing learner, loss, window and seed-pinned data order — so any FADO–Aranyani gap is the controller alone.

5. Results

Marginal DP — drift erodes fairness; controller helps on **abrupt** shift

Dataset	Model	Acc $\uparrow$	DP $\downarrow$
Folktables	FADO	0.8033	0.0284^{**}
	Aranyani	0.8053	0.0299
	ARF	0.7447	0.0722
	RFR	0.7780	0.0733
COMPAS	FADO	0.6434	0.0611^{***}
	Aranyani	0.6451	0.0679
	ARF	0.6313	0.0808
	RFR	0.6404	0.0813

Reading the table

RQ1. Both external baselines carry $\sim 2\times$ FADO's DP on both regimes (all $p<0.001$).

Regime-dependent. On abrupt COMPAS the controller cuts DP by 0.68pp ($p<0.001$).

On gradual Folktables it ties the base learner — a smooth shift needs no spike.

*Bold = best per metric within dataset; ^{**} $p<0.01$, ^{***} $p<0.001$ (paired Wilcoxon, Holm) vs every baseline.*

6. Intersectional Fairness

Marginal parity is **provably gameable**

Enforcing parity on **each attribute separately** is defeatable (Kearns et al., 2018): a model can be perfectly fair on sex *and* on race, yet badly unfair on a **joint** subgroup — fairness gerrymandering.

FADO optimises parity over the **Cartesian-product subgroups**. The regulariser already supports arbitrary $K \geq 2$ — the learner needed no change.

$$a = 2a_1 + a_2 \in \{0, 1, 2, 3\}$$

One 4-valued composite index replaces two binary axes.

Four joint cells

$a_1=0, a_2=0$	$a_1=0, a_2=1$
$a_1=1, a_2=0$	$a_1=1, a_2=1$

Intersectional grouping surfaces the **controller's contribution**

Model	Acc $\uparrow$	DP $\downarrow$	DP sex $\downarrow$	DP race $\downarrow$
FADO	0.7979	0.0688	0.0234	0.0243
Aranyani	0.8005	0.0727	0.0283	0.0294
ARF	0.7446	0.1497	0.0735	0.0620
RFR	0.7779	0.1427	0.0724	0.0591

DP = joint violation over the four cells (optimised); DP sex / DP race = marginal diagnostics. Bold = best mean.

The controller now measurably wins

vs the controller-free forest: DP -0.38pp , significant under **both** Wilcoxon ($p < 0.001$) and Welch ($p < 0.01$) — where the marginal view was only a tie.

Gerrymandering, confirmed

Intersectional-to-worst-marginal DP ratio ≈ 2.8 (FADO), 2.5 (Aranyani), ~ 2.0 (ARF/RFR): the fairest learners hide the largest joint gap.

7. Takeaways & Next Steps

What we learned — and what's next

Takeaways

RQ1. Drift is a fairness problem — it inflates the DP of accuracy-only and static-fair baselines alike.

RQ2. An adaptive fair learner keeps both utility and fairness; the controller earns its keep on abrupt shift.

Design. The fairness objective must match the harm: node-level DP + a joint group index makes the penalty cheap and the controller's gain measurable.

Next

Controller ablation — isolate each phase; the most targetable point.

Optimise EO directly (wire the existing EO regulariser into the gradient).

Finish the COMPAS intersectional sweep and re-verify gerrymandering.

Fairness that adapts

A drift-aware controller keeps an intersectionally-fair oblique forest fair **as the distribution moves.**

Thank you — questions & discussion

Tiago Silva · CISUC Data-Centric AI Lab · FADO (submitted to ICDM 2026)

Is a forest more interpretable than a DNN?

Sense of “interpretable”	Axis-aligned tree	Soft oblique forest	Deep NN
Simulatability – trace one prediction	Yes	Partial	No
Decomposability – parts carry meaning	Yes	Yes	No
Post-hoc – SHAP / saliency	Yes	Yes	Yes

Oblique (linear-combination splits) and **soft** routing (probabilistic paths) trade rule-level transparency for expressiveness + differentiability — so it is *not* as readable as a plain if-then tree.

What FADO actually leans on is decomposability: fairness is enforced and audited **per routing node** — a structure a monolithic network simply does not have.

Why backpropagation is the **enabling mechanism**

Reverse-mode autodiff: gradients w.r.t. **all** parameters in one backward pass, at $\sim$ the cost of one forward pass (finite differences would cost one pass *per* parameter).

-
- 1. Makes “soft” pay off.** Train all oblique hyperplanes *jointly* by gradient descent, instead of greedy CART splits.
 - 2. Enables streaming.** Cheap per-sample SGD step — the model adapts incrementally as the stream flows; a greedy tree cannot update per sample.
 - 3. Lets fairness enter training.** $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{fair}}$: backprop carries *both* the accuracy and the fairness gradient to every node.
 - 4. The controller needs a gradient loop.** It modulates the learning rate η and temperature τ — knobs that only exist under gradient training.
 - 5. Keeps the step cheap.** Node-local fairness $\Rightarrow$ the fairness gradient is $O(\text{nodes})$, fast enough for real-time streams.